

The Credentials Arms Race: Understanding Immigrant Occupational Mismatch in the U.S. Labor Market

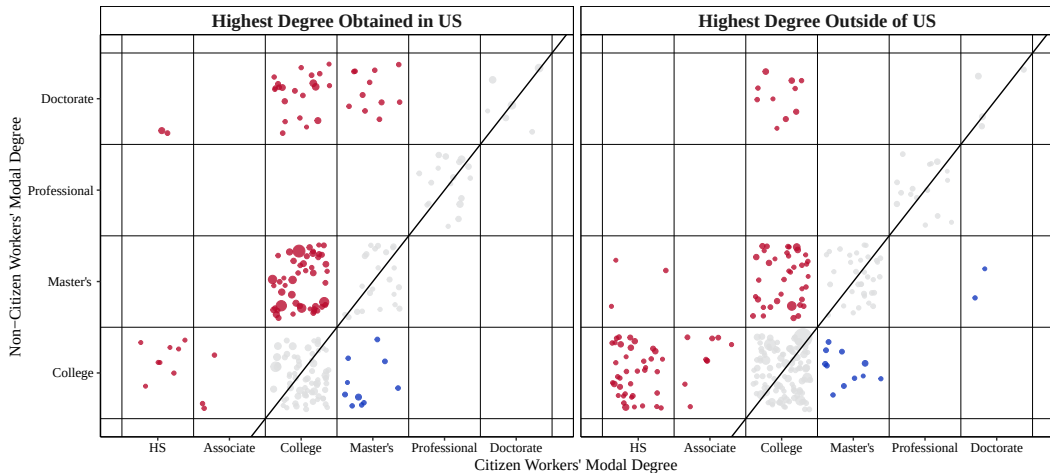
Willy Chen

Grand River Workshop
May. 20, 2026



Motivation: The Mismatch Phenomenon

Figure 1: Educational Mismatch of Non-Citizens, ACS 2010-2019



Core Objectives

- ① Document Sorting: Map empirical patterns of vertical and horizontal credentials mismatch using realized job-frequency distributions.
- ② Isolate Equilibrium Frictions: Extend the productive signaling framework to a continuum of worker types to distinguish asymmetric information from persistent labor market barriers.
- ③ Quantify Earnings Gaps: Structurally decompose the immigrant-native wage gap into sorting inefficiencies versus differential returns to education signals.

Occupational Mismatch: Definition and Impact

- ① Language Proficiency and Occupational Assimilation (Beenstock et al., 2010; Chiswick et al., 2009, 2010; Collins et al., 2023; Imai et al., 2019).

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- ② Vertical Mismatch: Over-Education and Wage Penalty (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).

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- ③ Horizontal Mismatch: Unused Specialization (Cassidy, 2019; Chiswick et al., 2013; Hartog, 2000; Imai et al., 2019; Li et al., 2023; Weber et al., 2024).

- LinkedIn worker-level career and education history.
- LinkUp job titles with O*NET taxonomy.
- Occupation Information Network (O*NET).
- American Community Survey (ACS), 2005-2019 and 2021-2024.

LinkedIn Profiles

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- **Came to study:** First post-secondary degree is from a non-US institution and at least one post-secondary degree is from a US institution.
- **Came to work:** All post-secondary degree are from non-US institutions and has career spells in the US.

LinkedIn Panel

I construct a panel of individuals who either obtained a degree from a U.S. degree-granting post-secondary institution or held at least one job in the U.S. The sample spans 1970 – 2025 and includes ~790M person-year observations from ~54M (40M with status) individuals, averaging 14.5 years per person.

Table 1: Degree and Graduation Year Distribution

Part 1: By Terminal Degree					Part 2: Graduation Cohorts			
% Degree	LinkedIn			ACS (3.8M)	Cohort	US	CtS	CtW
	US (25.5M)	CtS (2.2M)	CtW (12.4M)					
Bachelor's	60.9	13.6	54.0	64.2	Pre '00	8.0	4.0	4.0
Master's	31.6	68.5	41.4	26.0	'00-'04	5.0	4.1	4.7
Professional	4.6	3.4	0.8	5.8	'05-'09	9.0	9.7	8.3
PhD	2.9	14.5	3.9	4.0	'10-'14	16.8	17.7	16.0
					'15-'19	22.6	25.3	25.6
					'20-'25	38.6	38.5	42.0

*US=All US Edu. CtS = Foreign edu. then US edu. CtW = No US edu.

Matching LinkedIn Job Titles to SOC's

Using 250 distinct job titles for each of the 1,016 SOC codes from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

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Map the job title to its SOC code. Output the code only. For gibberish, use 00-0000.00.

Instruction: Map title to SOC

Input: Vice President, company_name

Response: 11-1011.00

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Out of $\sim 8.6T$ employed job-year rows, 90% are recovered with a valid SOC code with an average confidence score of 0.87 out of 1.

Matching LinkedIn Job Titles to SOC's

Table 2: Example of Predicted SOC Codes using Preliminarily Fine-Tuned Llama-3.1

LinkedIn Title	O*NET SOC	Corresponding O*NET Title
director - field support	27-2012.05	Media Technical Directors/Managers
senior page designer and copy editor	27-3041.00	Editors
gflwn askdm	00-0000.00	NA
taylor mattis fellow	51-9195.05	Potters, Manufacturing
solar analyst intern	17-2199.11	Solar Energy Systems Engineers
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9032.00	Education Administrators, Kindergarten through Secondary
asset & reliability engineer	17-2199.00	NA
sales / acct manager	41-1011.00	First-Line Supervisors of Retail Sales Workers
attest intern	13-2099.00	NA
immersion associate	25-1194.00	Career/Technical Education Teachers, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers

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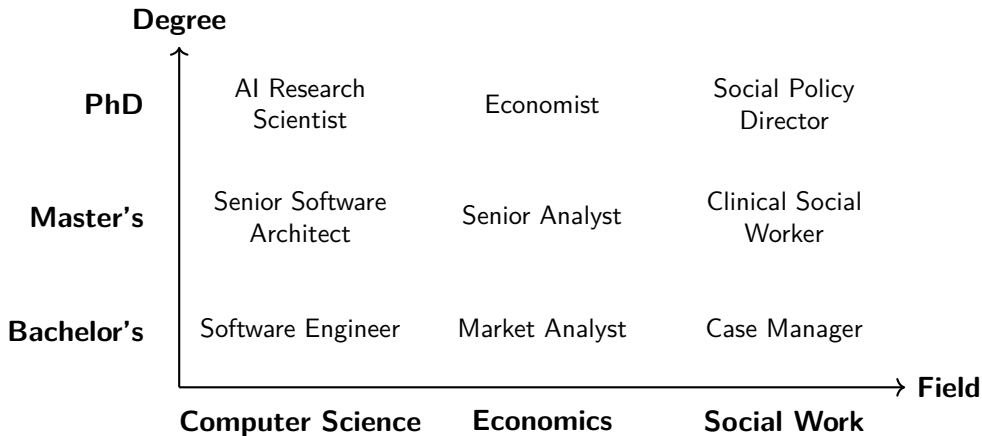
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網頁工程師	15-1299.08	Computer Systems Engineers/Architects
博士生	25-9044.00	Teaching Assistants, Postsecondary

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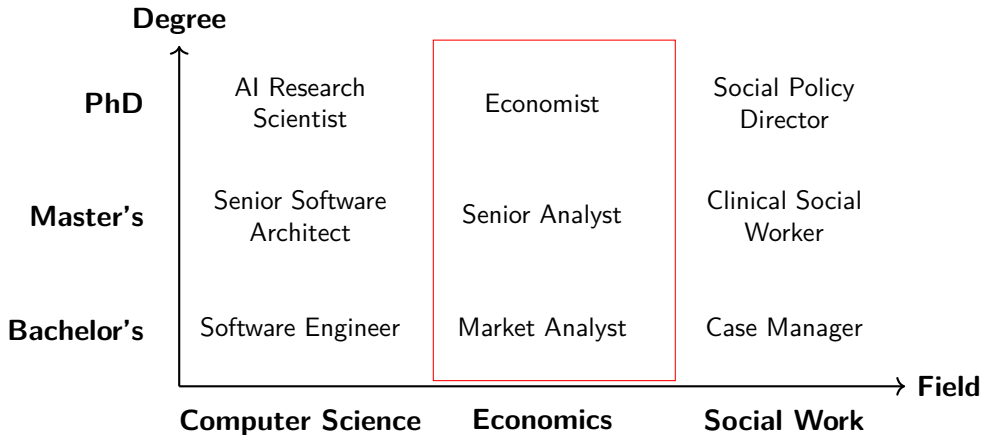
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網頁工程師	15-1299	Computer Systems Engineers/Architects
博士生	25-9044	Teaching Assistants, Postsecondary

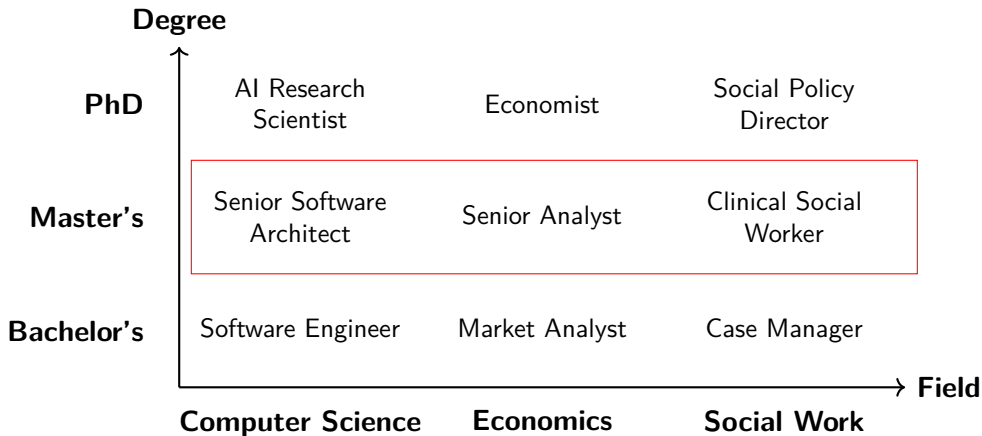
The Taxonomy of Occupational Mismatch



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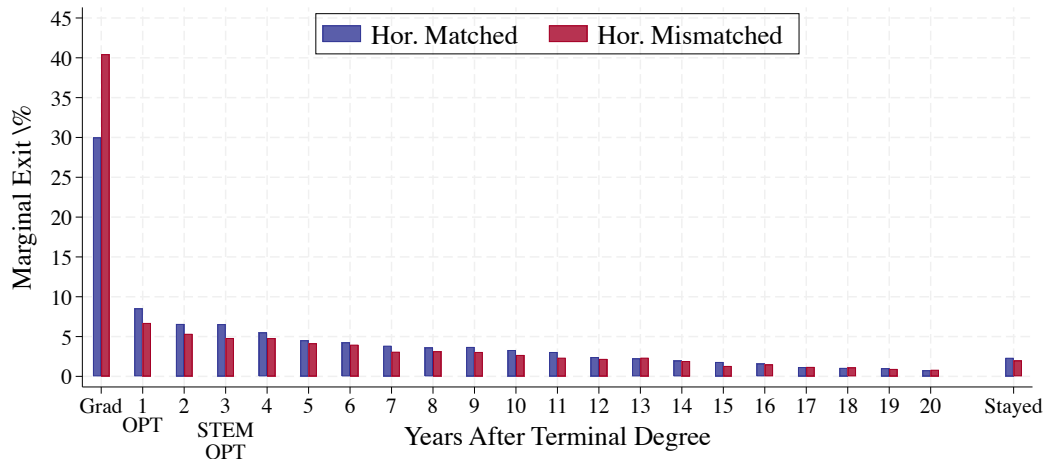
Defining Horizontal Mismatch

Figure 2: Distribution of First Jobs Among Most Common Jobs Within Degree-Field



Horizontal Mismatch and Exits

Figure 3: Marginal Exit Rates by First-Job Mismatch Status



Predictors of Occupational Mismatch

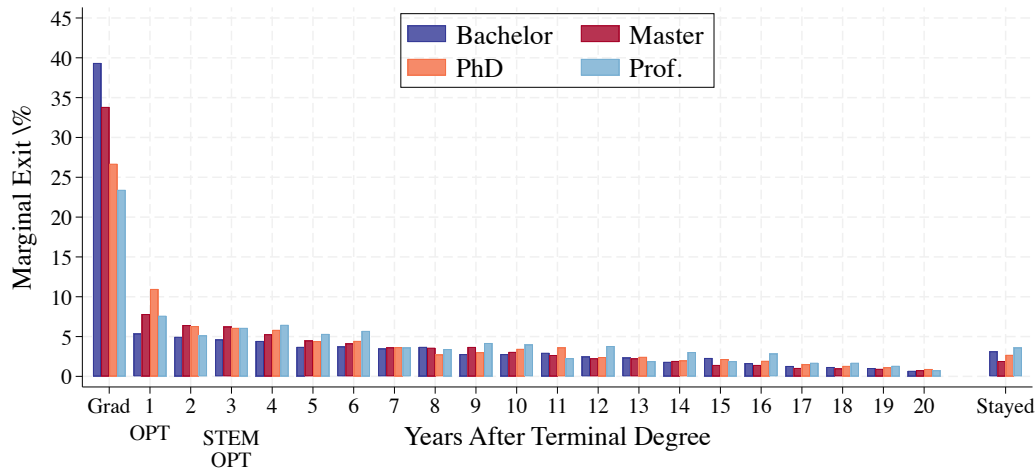
Table 3: Probability of First Job Any Mismatch for Immigrants

	Pr(Any Mismatch)			
Came to Study	-0.062*** (0.009)	-0.068*** (0.013)	-0.046** (0.016)	-0.053** (0.018)
Masters	0.228*** (0.002)	0.254*** (0.007)	0.334*** (0.008)	0.345*** (0.009)
PhD	0.186*** (0.006)	0.185*** (0.013)	0.280*** (0.016)	0.265*** (0.017)
Professional	-0.079*** (0.005)	-0.085*** (0.013)	0.039** (0.014)	0.069*** (0.016)
Came to Study × Masters	0.118*** (0.010)	0.104*** (0.014)	0.140*** (0.017)	0.147*** (0.019)
Came to Study × PhD	0.105*** (0.013)	0.109*** (0.017)	0.032 (0.021)	0.035 (0.023)
Came to Study × Professional	0.147*** (0.021)	0.144*** (0.026)	0.117*** (0.028)	0.113*** (0.032)
N	268,531	247,610	156,257	127,498
Additional Controls	Cohort FE	Degree × School Tier	O*NET Job Char.	Firm Utility.

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Exits by Degree

Figure 4: Marginal Exit Rates by Degree



Occupational Mismatch: Theory and Identification

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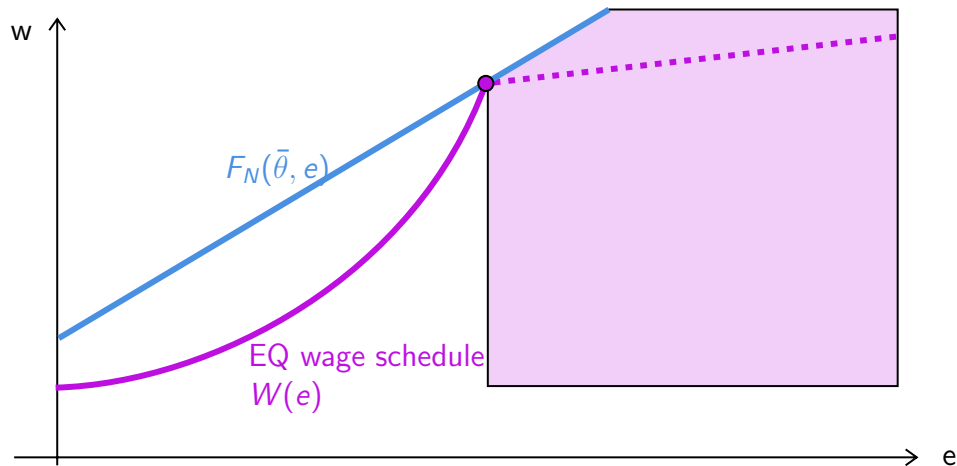
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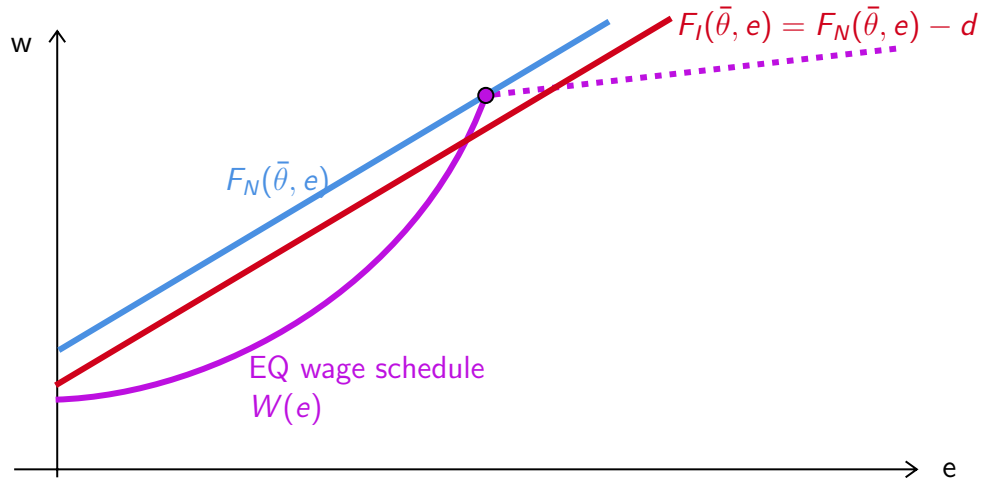
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- ⑤ Non-pecuniary benefits in sorting: (Lachowska et al., 2023; Sorkin, 2018).
- ⑥ Challenges with Immigrants Workers (Bhalothia et al., 2025; Bound et al., 2015; Khanna, 2025; Khanna and Morales, 2024; Lubotsky, 2007).

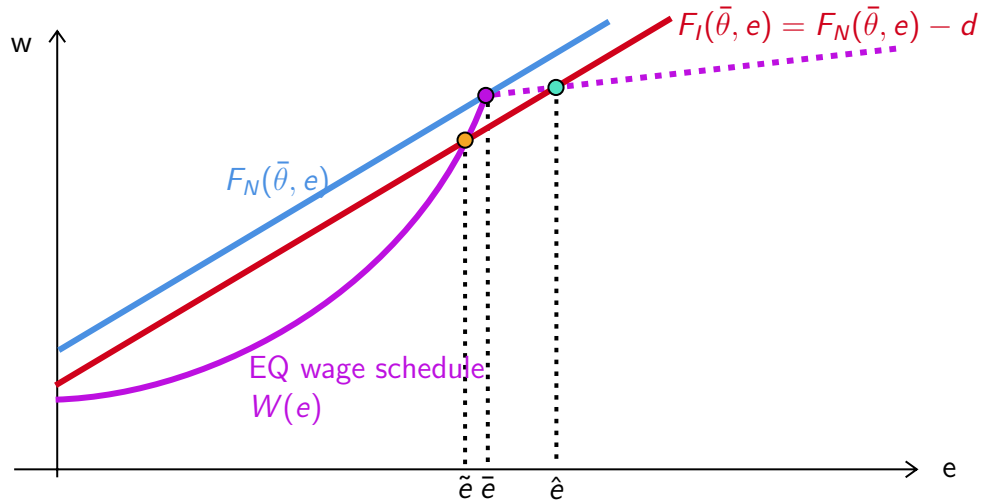
Model: Productive Signals with Continuous Types



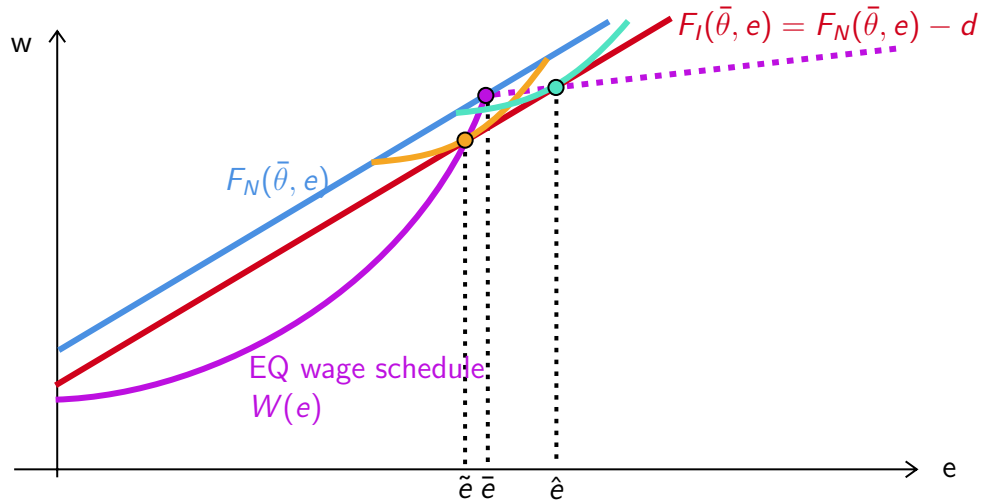
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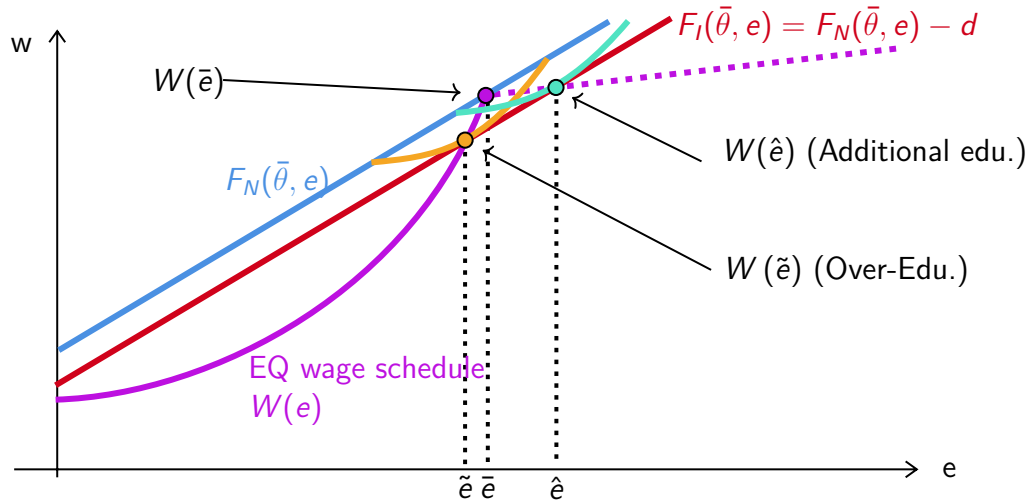
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$$d = W(\bar{e}) - \left[W(\tilde{e}) + \frac{W(\hat{e}) - W(\tilde{e})}{\hat{e} - \tilde{e}} (\bar{e} - \tilde{e}) \right].$$

At initial placement, $d = d^p + d^t$. Over time, $d \rightarrow d^p$. By observing within-cohort d over time, I can bound d^p and d^t .

Constructing Education Signal $\hat{e}$

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$$\text{score}_s = 0.65 \times \text{Std. SAT/ACT Scores} + 0.35 \times \text{Std. Rejection Rate}$$

Applying the prestige proxy to each degree-level requires strong Independence of Irrelevant Alternatives (IIA) assumption.

A reasonable alternative is to bin schools by selectivity ranking to remain agnostic about the added returns to signal in each tier.

Constructing Education Signal $\hat{e}$

Table 4: Selectivity Bins

Bin	1	2	3	4	5	6	7	8	9
Selectivity Ranking	NA	Bottom 25%	Top 75%	Top 50%	Top 33%	Top 20%	Top 10%	Top 5%	Top 1%

I calculate the education signal for each degree-school pair using $\mathbb{I}$ from the first job after the last degree in a worker's history:

$$\mathbb{I}_i = \beta_0 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \delta_d \cdot \mathbb{1}\{Deg_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \sum_{b=1}^{10} \beta_{d,b} \cdot \mathbb{1}\{Deg_i = d\} \cdot \mathbb{1}\{Bin_i = b\} + \varepsilon$$

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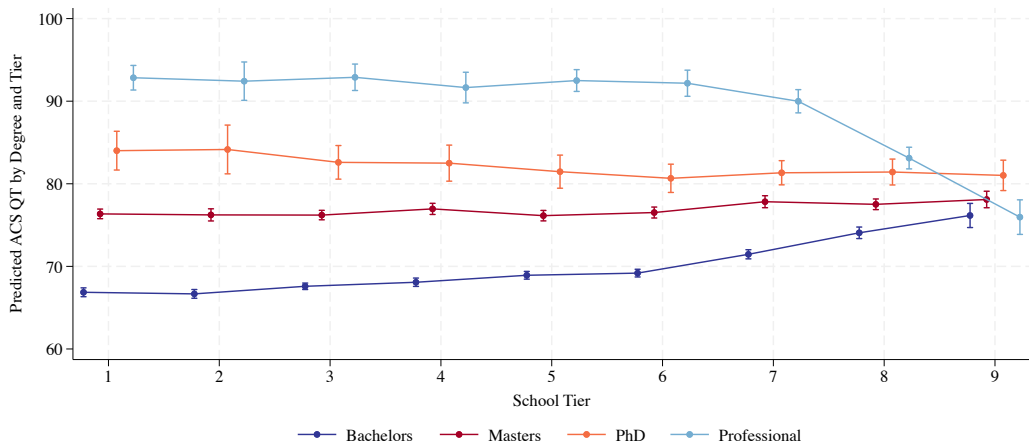
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Natural candidates of $\mathbb{I}$: Occupational quantiles and occupational prestige.

Selection Issues at High Tier Schools



Network Job Characteristics via Realized Matches

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 - SOC Education Source Heterogeneity (DUH/HHI SOC)

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② Importance of Certification vs. Typical On-the-Job Training [Graphs](#)

ACS Job Characteristics

I use ACS to compute two measures using only workers aged 25-64 who are employed, not self-employed, not in the military, and not in school:

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- ① Within-occupation distributions of educational attainment for citizen workers (2010-2019).
- ② SOC-MSA-Year level wage at the 10th, 25th, 50th, 75th, and 90th percentiles in 2015 dollars.

Equilibrium Wage Schedule

Table 5: Average Marginal Effects of Education Signals on Job-Level Median Wages, All US Education

In(Edu Signal)	0.158*** (0.002)	0.145*** (0.003)	0.153*** (0.004)	0.147*** (0.004)	0.191*** (0.004)	0.148*** (0.004)	0.191*** (0.004)
Years since Degree	0.007*** (0.000)	0.007*** (0.000)	0.007*** (0.000)	0.008*** (0.000)	0.008*** (0.000)	0.008*** (0.000)	0.008*** (0.000)
Major Generality		-0.004 (0.002)	0.003 (0.003)	0.010** (0.003)	0.008* (0.003)	0.011*** (0.003)	0.009** (0.003)
SOC Generality		-0.032*** (0.002)	0.059*** (0.003)	0.046*** (0.003)	0.060*** (0.003)	0.046*** (0.003)	0.061*** (0.003)
Firm Util.				0.006*** (0.000)	0.006*** (0.000)	0.006*** (0.000)	0.006*** (0.000)
SOC Util.				0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)
ACS Over Edu.					-0.014*** (0.001)		-0.014*** (0.001)
Hor. Mismatched						-0.001 (0.001)	-0.001 (0.001)
Observations	939,374	845,050	576,680	501,749	501,749	501,749	501,749
Additional Controls	Cohort FE		ONET Char.	Churn Rates			

Notes: Standard errors in parentheses. ONET characteristics include cognitive and interpersonal competence, attention and coordination, practical job capabilities, indicators of time for typical on-the-job training, and importance of certification. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Oaxaca-Blinder: First-Job Wage Gap Dynamics by Educational

Residualized Log(SOC-MSA-Year Median Wage)

	(1) All	(2) Bachelor	(3) Master	(4) PhD & Prof.
Came-to-Study	0.050***	-0.006	0.062***	0.026**
All US Edu.	-0.001	-0.006***	0.005***	0.023***
CtS - US Gap	0.050***	-0.001	0.057***	0.003
— <i>Explained</i>	0.023***	-0.006*	0.022***	0.009
— <i>Unexplained</i>	0.027***	0.005	0.035***	-0.006

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Key Compositional Shifts (Explained)

$$(\bar{X}_{CTS} - \bar{X}_{US})\beta_X$$

- **Master's Single-Channel Sorting:** For Master's holders, field-specific signal intensity ($\ln(\text{Edu. Signal})$) is the exclusive composition driver of sorting into higher-median-wage jobs (+0.011***). Broad task tracks (SOC Generality) and certification importance are completely silent at this tier.
- **Advanced Tier Equilibrium:** In contrast, the Doctorate/Professional tier's zero explained gap (0.009) is a high-stakes compositional tug-of-war. Group 1's strong baseline field signal advantage (+0.034***) is completely neutralized by sorting into highly specialized, non-general occupations (SOC Generality at -0.017***) and intense over-education layout penalties.

Key Structural Shifts (Unexplained)

$$\bar{X}_{CTS}(\beta_{X,CTS} - \beta_X) - \bar{X}_{US}(\beta_{X,US} - \beta_X)$$

- **Master's Sorting Paradox:** The significant unexplained sorting gap for Master's holders (0.035***) masks extreme structural forces. At the baseline, over-educated Master's holders work low-median-wage occupations (-9.328*). However, this penalty is entirely neutralized if they possess a strong educational signal ($0E \times \ln(\text{Signal})$ at +9.339*).
- **Advanced Certification Sorting Barriers:** In fields where professional certifications are important, the Came-to-Study Doctoral degree holders face an unexplained sorting penalty (-0.021**), structurally steering them into lower-median-wage occupations relative to the All-US-Edu group.

Questions?

Thank You!

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General Separating Equilibrium

The general model follows the work of (Mailath, 1987).

Each worker chooses their education according to $\sigma : \Theta \rightarrow \mathcal{E}$.

Observing e , the firm updates belief about the worker's type $\hat{\Theta}(e) \subseteq \Theta$.

In a full **separating** equilibrium, σ is a one-to-one function, i.e.,

$$\hat{\Theta}(e) = \{\sigma^{-1}(e)\}, \forall e \in \mathcal{E} \quad (1)$$

and $\sigma(\theta)$ must satisfy *strict incentive compatibility*:

$$\{\sigma(\theta)\} = \underset{e}{\operatorname{argmax}} U(\theta, \sigma^{-1}(e), e), \forall \theta \in \Theta \quad (2)$$

Regularity Conditions

- ① (Smoothness) $U(\theta, \hat{\theta}, e)$ is twice differentiable on $\Theta^2 \times \mathcal{E}$.
- ② (Belief Monotonicity) $\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e) > 0$.
- ③ (Type monotonicity) $\frac{\partial^2}{\partial \theta \partial e} U(\theta, \hat{\theta}, e) > 0$
- ④ (“Strict” Quasi-Concavity) $\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) = 0$ has a unique solution in e , denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{\operatorname{argmax}} U(\theta, \hat{\theta}, e)$, and $\left. \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \right|_{e=\phi(\theta)} < 0$.
- ⑤ (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \Rightarrow \left| \frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) \right| > k.$$

Additionally, there are two niceness conditions: Initial Value (IV) and Single Crossing (SC).

Characterization of the Solution

Mailath (1987) shows that

If conditions 1-5 are satisfied, then a strictly increasing function σ , continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

- ① σ solves
$$\frac{d\sigma}{d\theta} = - \frac{\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e)}{\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)}$$
- ② when $\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e) > 0$, $\frac{\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)}{\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e)}$ is a strictly increasing function of θ .

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

Characterization of the Solution

Mailath (1987) shows that

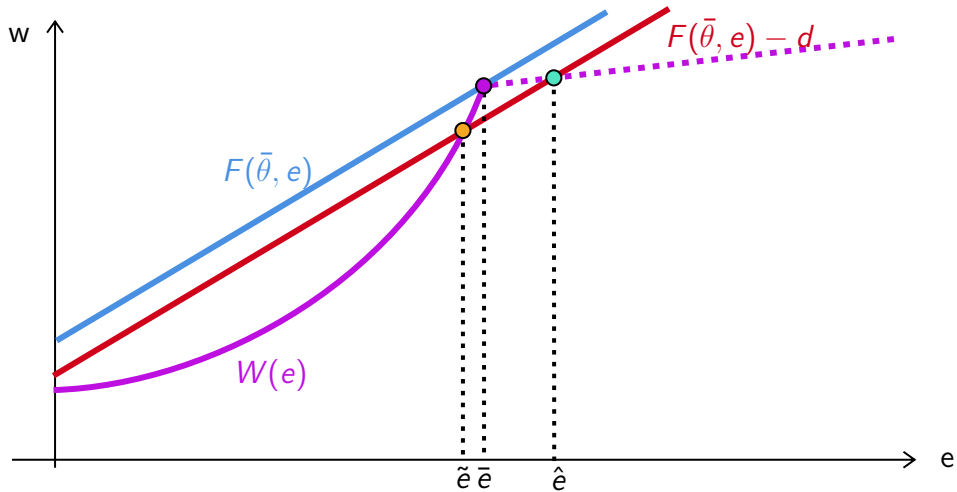
If conditions 1-5 are satisfied, then a strictly increasing function σ , continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

- ① σ solves
$$\frac{d\sigma}{d\theta} = - \frac{\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e)}{\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)}$$
- ② when $\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e) > 0$, $\frac{\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)}{\frac{\partial}{\partial \hat{\theta}} U(\theta, \hat{\theta}, e)}$ is a strictly increasing function of θ .

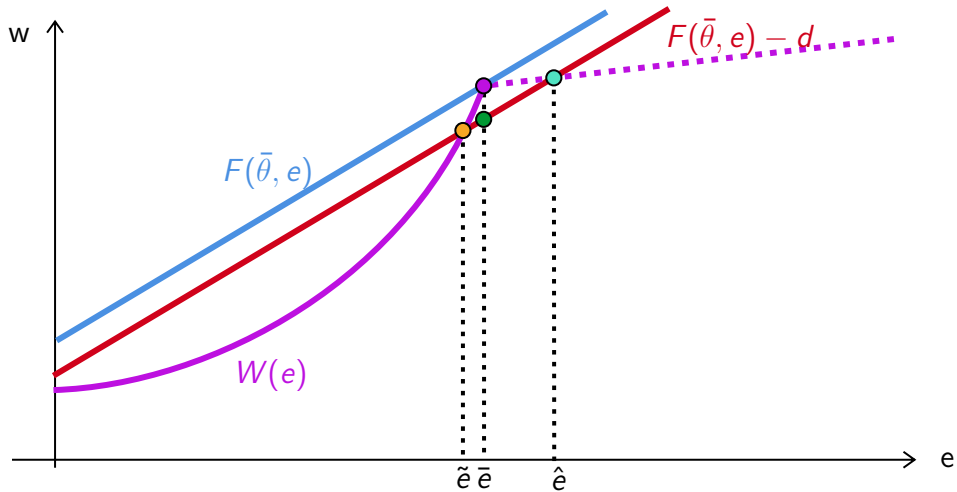
Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

In other words, there is an unique full separating equilibrium in the natives' labor market given the assumptions in the basic model environment.

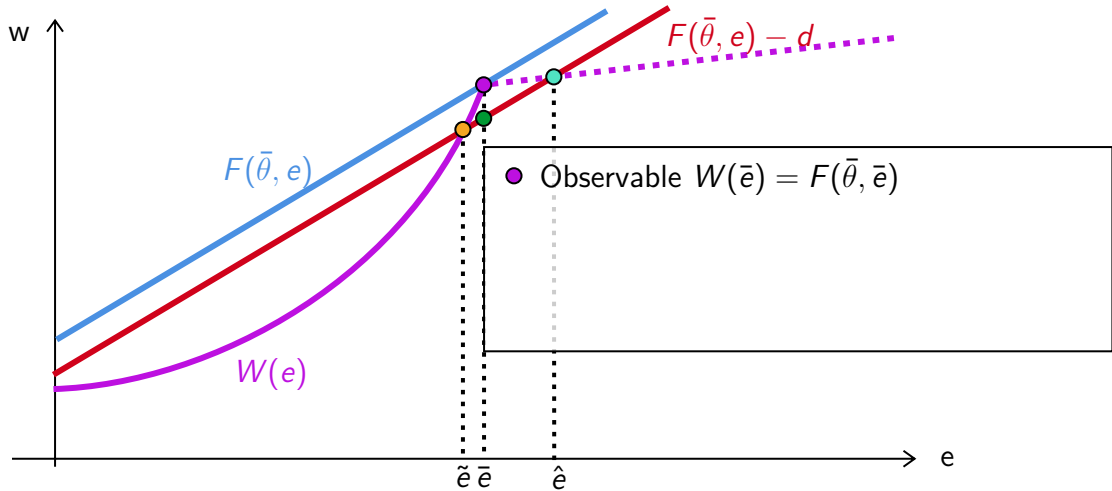
New Separating Equilibrium



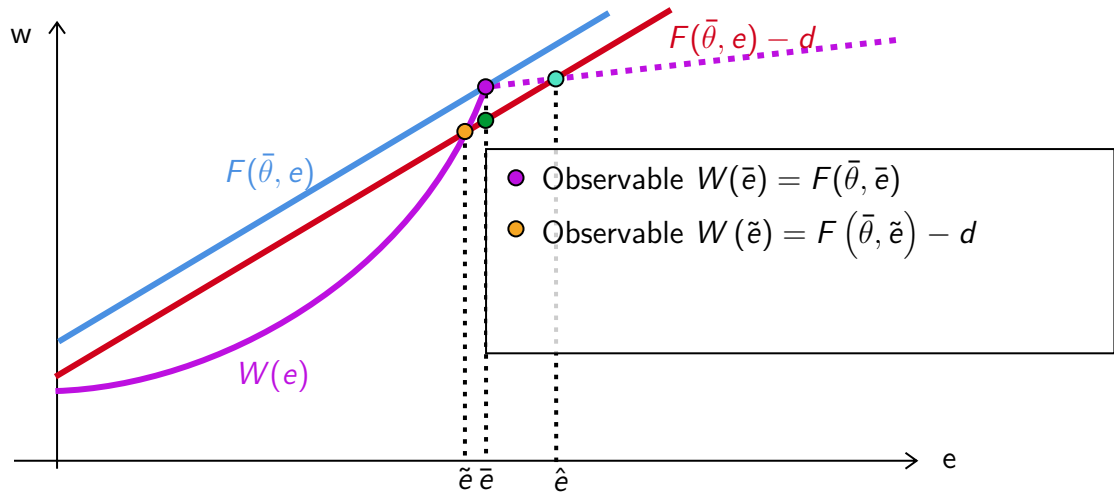
New Separating Equilibrium



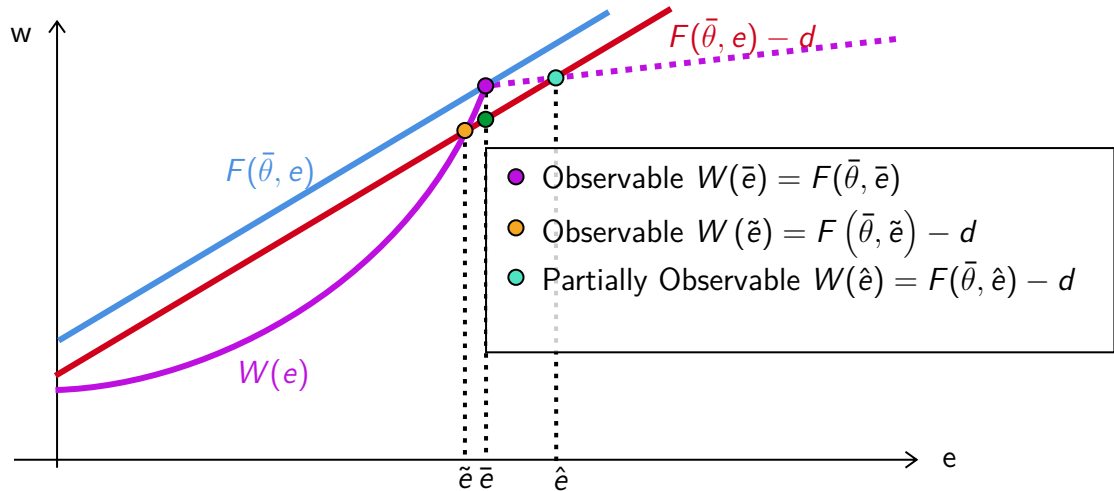
New Separating Equilibrium



New Separating Equilibrium



New Separating Equilibrium



New Separating Equilibrium

